BEFORE THE
STATE OF NEW YORK
PUBLIC SERVICE COMMISSION

In the Matter of

Consolidated Edison Company of New York, Inc.

Cases 25-E-0072 25-G-0073

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Prepared Testimony of:

Ben Johnson, Ph.D.
Consulting Economist
Ben Johnson Associates, Inc.
606 Bayport Way
Longboat Key, FL 34228

I. INTRODUCTION

Q. Would you please state your name and business address?

A. My name is Ben Johnson. My address is 606 Bayport Way, Longboat Key, FL 34228.

Q. Who is sponsoring your testimony in these proceedings?

A. A bipartisan group of 40 Westchester County cities, towns, and villages. I will refer to these communities as the "Westchester Municipal Consortium," or the "Consortium" for brevity.

Q. By whom are you employed, and what is your professional background?

A. I am a Consulting Economist and President of Ben Johnson Associates, Inc., an economic research firm I founded in 1979 that specializes in public utility regulation. I received a Bachelor of Arts in Economics from the University of South Florida and a Master of Science in Economics and Doctor of Philosophy in Economics from Florida State University.

Throughout my career, I have specialized in the public utility industry. I have been involved in more than 400 regulatory dockets involving electric, natural gas, telecommunications, and other utilities. Most of this work has been performed on behalf of utility regulatory commissions, consumer advocates, and other government agencies.

My firm's clients have also included large industrial customers (primarily in the Pacific Northwest), independent power producers and other firms that compete with regulated utilities, the United States Department of Justice, the Florida Municipal Electric Association, the AARP (formerly the American Association of Retired Persons), and the North Carolina Sustainable Energy Association.

Q. Have you previously testified before the Commission?

A. Yes. I submitted testimony in Cases 13-E-0030

and 13-G-0031 (Con Edison), 14-E-0493 and 14-G-0494 (Orange & Rockland), 15-E-0283, 15-G-0284, 15-E-0285 and 15-G-0286 (NYSEG and RG&E), 16-G-0058 and 16-G-0059 (KEDLI and KEDNY), 16-E-0060 and 16-G-0061 (Con Edison); 16-G-0257 (National Fuel); 17-E-0238 and 17-G-0239 (Niagara Mohawk), 17-E-0459 and 17-G-0460 (Central Hudson); and 19-E-0378, 19-G-0379, 19-E-0380, and 19-G-0381 (New York State Electric & Gas and Rochester Gas & Electric); 21-G-0073 and 21-E-0074 (Orange and Rockland), 22-E-0064 and 22-G-0065 (Con Edison), and 22-E-0317, 22-G-0318, 22-E-0319, and 22-G-0320 (New York State Electric & Gas and Rochester Gas & Electric), and most recently in 24-E-0060 & 24-G-0061 (Orange & Rockland).

Q. Have you previously provided expert testimony in other jurisdictions?

A. Yes. I have testified (or submitted written testimony) on more than 250 occasions throughout North America. More specifically, I have provided expert testimony to state courts, federal regulatory agencies, Canadian provinces,

and the District of Columbia, but most of my professional experience has involved expert testimony provided to approximately 35 state regulatory commissions and agencies.

Q. What is the scope of this testimony?

A. I was asked to review the rate increases requested by Con Edison (the "Company"), discuss the impact of high electric and gas rates, and provide recommendations concerning the proposed rates.

My testimony has five sections. In this first section, I introduce myself and deal with a few preliminary matters.

In the second section, I compare the electric rates paid by residential and commercial customers in the 40 cities, towns, and villages (municipalities) sponsoring my testimony relative to various benchmarks. In this section, I identify some of the reasons why customers in Westchester County are paying much higher rates

than the customers in similar communities in other parts of New York and in other states.

In the third section, I discuss the steps typically followed in designing a utility's rates, the role public policy plays in that process, and the use of cost studies to support that effort.

I focus on the Company's rate design proposals in the fourth section. I highlight some controversial methodologies the Company employs and discuss alternative approaches that have been widely used in other regulatory proceedings. I also discuss some of the judgments made by the Company in developing the ECOS studies in this proceeding, particularly the decision to separate SC3 customers depending on the number of residential units on a single meter.

Finally, in the fifth section, I offer some recommendations and illustrate these

recommendations with specific suggestions concerning the proposed electric customer charges, gas minimum charges, and the flat, inclining, and declining block rates in certain tariffs.

Q. Have you prepared any exhibits to be filed with your testimony?

A. Yes. I have prepared five exhibits. Each exhibit has multiple pages. Exhibit__(WMC-1) comparatively examines the Company's rate levels. Schedule 1 focuses on electric rate increases outlined in Joint Proposals in the Company's 2016, 2019, and 2022 rate proceedings, while Schedule 2 provides similar data for the gas rates. Both schedules compare the recent price increases to the overall rate of inflation. Schedule 3 provides the inflation data used in developing those comparisons. Schedule 4 compares the Company's rates to those charged by other utilities.

Exhibit__(WMC-2) and Exhibit__(WMC-3) are demonstrative exhibits that clarify and illustrate portions of my testimony. Exhibit__(WMC-2) focuses on electric delivery rates while Exhibit__(WMC-3) focuses on gas delivery rates. Both exhibits have similar schedules. Schedule 1 provides class-specific customer cost estimates using some approaches I have seen applied in other proceedings. Schedule 2 illustrates my rate design recommendations assuming the Company's proposed revenue requirement is approved as filed. Schedule 3 provides estimated bill impacts corresponding to these illustrative rates. Schedule 4 illustrates my rate design recommendations assuming the Commission adopts a lower revenue requirement. Schedule 5 provides estimated bill impacts corresponding to the illustrative rates developed using this assumed lower revenue requirement.

Exhibit__(WMC-4) contains the Company's responses to Information Requests (IRs) that I

relied upon in preparing this testimony. Exhibit__(WMC-5) contains a list of the communities that comprise the Municipal Consortium. Table 2 provides some information about each community, confirming the diversity of their populations, income levels, energy usage, and other characteristics.

II. CONSORTIUM CONCERNS

Q. Why did 40 cities, towns, and villages in Westchester County decide to participate in this proceeding?

A. I cannot get into the minds of the many town administrators, mayors, city council members, and other public officials who decided to move forward with this unprecedented intervention, but several things are apparent.

First, this intervention is quite unusual. It may be unprecedented. I am familiar with all the Con Edison rate cases since Cases 13-E-0030 and 13-G-0031 (filed in January 2013). The closest

analogous intervention efforts have been those of New York City. Other intervenors have typically included environmental groups (like the Environmental Defense Fund, the Alliance for a Green Economy, and the Natural Resources Defense Council) and groups of large energy users (like Consumer Power Advocates, the Metropolitan Transportation Authority, and the New York Energy Consumers Council). I do not recall any other Con Edison proceedings in which a group of smaller cities, villages, and towns intervened.

Second, this intervention directly responds to Con Edison's high rates and the threat of even higher rates being adopted by the Commission. The consortium was formed in large part due to concerns about the financial strain imposed on their residents and businesses by high utility bills and the potential for additional economic impacts if the proposed increases are approved.

Third, the Consortium is not seeking a subsidy analogous to that provided to low-income customers. Yet, the underlying concern is similar to the reasons why the Commission has gone to great lengths to help keep utility service affordable for low-income customers. Although Westchester County is fortunate to have some wealthy residents, not everyone fits that category. These communities have many small businesses and moderate-income residential customers and senior citizens struggling to squeeze high utility bills into their limited budgets. Only the wealthiest citizens and the most profitable businesses in Westchester County can shrug off ever-mounting utility rates that significantly exceed those charged by utilities in similar small communities in Rockland County, Putnam County, Orange County, Nassau County, and Dutchess County. Even businesses that can afford increases may see diminished profits and look at alternative, less expensive locations outside Westchester in the future.

The President of the Westchester Municipal Officials Association (WMOA) was quoted in a press release announcing the formation of the consortium, saying, "Our communities must stand together to protect our residents and businesses from these excessive rate hikes." By joining forces and intervening in these proceedings, these municipalities seek to ensure that the Commission is aware of the "financial strain on residents and businesses throughout Westchester County." The Consortium is not seeking special treatment or a subsidy. It wants the Commission to pay attention to the difficult situation faced by their constituents, and to ensure that any approved rate changes are just and reasonable.

Q. Are there specific issues of particular concern to the Consortium members?

A. Yes. I was asked to look at the economic and geographic circumstances in Westchester County to address concerns regarding the adverse impact of high utility rates on the residents and

businesses in their communities. These issues have not been thoroughly investigated or discussed in any recent Con Edison proceedings.

Q. Can you be more specific?

A. Yes. These smaller communities have geographic characteristics and economic circumstances that differ from those in New York City (the "City"). Anyone who has experienced the City's vibrancy knows it is a very special place, yet we may fail to realize just how unusual it is. The City has a population of more than 8 million with a density of more than 29,000 people per square mile. Los Angeles, the second largest city in the United States, has less than half that population, and less than one-third the population density (about 8,000 people per square mile).

The most densely populated areas in Westchester County (communities like Mount Vernon and Port Chester) have characteristics that overlap those of the least densely populated parts of New York

City, like Staten Island (about 8,000 per square mile). Despite this overlap, it is fair to say that the situation in the average community in Westchester County is dramatically different than the situation in most areas within the City. This broad generalization also applies to many other variables, but it is sufficient to look at one elementary statistic: population density. The overall average density of Westchester County was approximately 2,300 per square mile in 2020. A typical community like **Rye City had a population density** of 2,800 per square mile. This is less than half the density of Staten Island (8,300 per square mile), approximately one-tenth the density of Brooklyn (36,600 per square mile) and the Bronx (33,200 per square mile). It is **less than one-twentieth the density of Manhattan**, which has 71,100 people per square mile.

Q. Are differing geographic and economic characteristics relevant to this proceeding?

A. Yes. The Consortium is asking the Commission to

set just and reasonable rates for their communities. To achieve that goal, the Commission should consider the differences in circumstances facing small communities in Westchester County relative to those in a world-class urban center, like New York City.

While exceptions can arise, as a general rule, the economics of providing utility service follow some clear patterns. The cost of delivering electricity and natural gas to a typical customer in a small town is less than what is required to provide that commodity in a highly congested, densely populated urban area. Costs in small municipalities also tend to be less than the cost of providing service in sparsely populated rural areas. The cost per mile to install facilities in a sparsely populated rural area is comparable, or slightly less. However, that is not the only consideration. Customers in rural areas tend to be widely scattered, with miles upon miles of roads running through forests and farmlands

before you reach the next house, farm, or business.

On balance, the cost of providing utility distribution service tends to be lower in small cities and towns relative to densely populated cities or sparsely populated rural areas with very few customers per mile of distribution line.

The Company's tariffs consider differences in voltage level, power factor, and peak usage characteristics. Yet, differences in the cost of providing service to highly congested areas are not reflected in the Company’s revenue allocation or rate design process.

Having looked at an array of readily available data and relying on my experience as an economist specializing in regulated utilities, I am confident that significant cost differences exist within Con Edison’s service area. It costs more to deliver energy to congested areas in

Manhattan than to less crowded areas in Queens or Staten Island.

Cost differences associated with congestion are compounded by differences attributable to the higher grade of service provided in areas like Manhattan. Generally, it costs more to provide extremely high reliability service to network customers using mostly underground facilities than to offer the more typical grade of service supplied to radial customers on overhead distribution lines.

Q. Does this discussion of cost differences have practical relevance in this proceeding?

A. Yes. The Commission has often found it appropriate to consider cost differences in defining customer classes, and it uses data concerning the cost characteristics of different types of customers to determine how the revenue requirement should be allocated to various classes or types of customers. The Commission has also accepted the use of cost data to help

determine the best way to design rates within each class. Underlying cost characteristics are given considerable weight, together with other relevant factors, such as the quality or grade of service and the impact of specific options on other important public policy goals, like promoting energy efficiency.

While the ratemaking process requires a careful balancing of multiple factors, the Commission has been emphasizing cost differences in recent years, as it has been provided with increasingly sophisticated cost analyses. This trend is best illustrated by recent efforts to understand the cost of providing standby service, which until recently was subject to a high degree of approximation, simplification, and cost averaging.

While there is no need to engage in anything as ambitious as those recent efforts, it would be highly appropriate to investigate differences in the cost of providing distribution service to

customers in the small cities and towns in Westchester County relative to the cost of providing service in other, more congested areas. That effort's primary, but not exclusive, focus should be differences in network systems relative to radial systems. There are stark, well-understood differences in engineering and performance of these systems. Ample data is available from the Company's continuing property records, accounting system, AMI system, and other internal data sources, making it feasible to understand and quantify these cost differences.

Q. If significant cost differences exist, how could this information be used in the rate-setting process?

A. The approach I recommend depends on the information obtained from a detailed investigation. However, we can anticipate that cost differences of sufficient magnitude and importance currently exist that will justify taking appropriate steps to better align

Westchester customer's rates with the actual cost of providing their service. One option is to provide a discount to radial customers due to the lower cost and quality of that grade of service. Another option is to maintain separate rate schedules for serving customers in the most costly locations. This would be somewhat analogous to the way Orange & Rockland customers are charged different rates than customers in New York City.

Ideally, rates would be closely aligned with the actual cost of providing service to largely homogenous groups of customers. Accordingly, the ideal approach will adjust rates to be more responsive to the nature, scope, and magnitude of the underlying cost differences. A well-designed study may reveal other less noticeable cost differences, in addition to the most obvious differences, like the distinction between network and radial systems. Those differences might or might not merit consideration during the ratemaking process,

depending on their magnitude, but an appropriate first step is to identify and quantify the cost differences.

It would be practical and consistent with well-founded cost recovery principles to adjust the Company's tariffs to reflect, to a reasonable extent, differences in the cost of providing service to network and radial customers. Among other options, this could easily be achieved by reorganizing how customers are sorted into customer classes, just as heating customers are sometimes grouped separately from other customers. However, this is not the only option. For instance, radial customers could remain within broader classes, but an appropriate discount would be offered (relative to network customers) to reflect the difference in cost and quality of service. Similarly, customers in unusually high-cost areas (like midtown Manhattan)could pay rates that reflect the higher cost of providing service in these unusually congested, high density locations,

relative to other, more typical network locations.

Q. Most solutions would add some complexity to the Company's tariffs. Do you think this would be justified?

A. Yes. The added complexity is not an insurmountable obstacle, given the wealth of information the Company's Automated Metering Information System provides and the associated billing capabilities. The Company knows whether a distribution area receives service primarily from a network or radial system. Further, the Company knows each customers' precise location on that system, the revenues they provide, and a wealth of other relevant details concerning their cost characteristics.

In my opinion, the cost differences I have been discussing are sufficiently significant they should be studied and quantified, so the Commission will be in a better position to take appropriate steps to ensure that all types of

customers within Westchester County are only paying their fair share of the revenue requirement and to ensure that differences in service quality are appropriately reflected in the tariffed rates. Network customers obtain significant benefits that are not available to radial customers, due to the added redundancy provided in a network design, and the increased reliability that occurs when system components are located underground. For instance, underground facilities are better protected from traffic accidents, ice storms, and other external environmental risks.

Q. From your perspective as an economist, are these differences important?

A. Yes. Con Edison's network systems are typically engineered with multiple sources of backup power, and multiple layers of redundancy are provided throughout the network. This costly configuration enables the Company to provide highly reliable delivery service to high-rise buildings in locations like Manhattan, which are

protected against failure even if multiple system components fail simultaneously. The transmission and distribution system in Manhattan is 100% underground per a post-Sandy report prepared by New York City. For all of New York City over eighty percent of the transmission and distribution cable is underground.In contrast, the majority of customers in Westchester county are connected to a radial system, and they are provided with a quality of service that is closer to the grade of service that is typically provided by Orange & Rockland and other electric utilities serving less densely populated locations like those that exist in most of Westchester County.

The extraordinary level of reliability provided to places like midtown Manhattan is highly valued by customers, but it comes at a cost. Con Edison’s network systems are more complex than its radial systems, and it can be harder to detect and more costly to resolve equipment issues in a network system, due to the

complexity of the system design and the difficulties involved with observing and testing system components located underground. This helps explain why interruptions may take longer to resolve when they do occur, and why Con Edison invests millions of dollars in extraordinary efforts to identify and resolve potential problems before they occur. For instance, it replaces less reliable cable types, outdated splices and joints, and obsolete underground switches in a costly effort to prevent problems before they occur.

Some indication of the difference in service quality can be gleaned by looking at the frequency of interruptions on Con Edison's network systems. The System Average Interruption Frequency Index (SAIFI) for Con Edison's network systems (excluding major storm events) as reported by the Staff of the Department of Public Service ("DPS Staff) in Case 24-E-0140 was .0117 in 2023. This metric compares to a statewide average SAIFI excluding Con Edison of

0.9400. To put these metrics into perspective, you could say that the Company's network customers face 1/80th the outage frequency risk as the statewide average customer in other service areas. This is a truly remarkable difference.

In contrast, the Company's radial systems had an SAIFI metric in 2023 of 0.3980. This compares favorably to the performance of the state's other utilities, but the achievement is not as remarkable. The DPS Staff reported other system-wide SAIFI metrics for 2023 that ranged from 0.7100 (Rochester Gas & Electric) to 1.2900 (New York State Electric and Gas).

The extraordinary lengths taken to provide extremely reliable service to Manhattan and other densely populated areas largely (but not entirely) explain why Con Edison was awarded the 2024 ReliabilityOne Award as the most reliable electric provider in the United States. The extraordinarily reliable service provided to

network customers also largely explains why the Company's service has been judged ten times more reliable than the national average. An average interruption that occurs on Con Edison's network systems lasts a little longer, but these interruptions in service are few and far between.

The Company's radial customers also receive more reliable service than customers in most other areas in the state, but the economic benefits are not as significant as those provided in Manhattan and Brooklyn. Instead of facing 1/80th of the statewide average outage frequency risk, Con Edison's radial customers are facing risks that are approximately half the statewide average. Stated another way, the outage frequency risk is about 34 times greater on the radial systems than on the network systems.

Another way of visualizing the difference in economic benefits is to compare Con Edison to its affiliate, Orange & Rockland. During 2023,

the Company's network customers faced 1/90th of the frequency risk borne by Orange and Rockland's customers. On the extremely rare occasions when a network customer experienced an interruption (despite all of the layers of redundancy and other costly measures taken to ensure that power is never interrupted), the problem required about six hours to resolve (on average). In contrast, the radial customers face one-third the frequency risk of Orange & Rockland's customers, and the problem requires less than two hours to resolve (about ten minutes longer than required on Orange & Rockland's system).

Q. Are these differences in service quality and corresponding differences in cost adequately reflected in the Company's rate design?

A. No. I am not aware of any benefit-cost studies or other efforts undertaken to analyze the economic impact of differences in service quality provided to Con Edison's radial customers relative to its network customers or

its affiliates' radial customers. Nor am I aware of any effort by the Commission to evaluate whether, and to what extent, this difference in economic benefits should be reflected in a corresponding difference in the rates paid by radial customers relative to those paid by network customers.

While these may be questions of first impression for the Commission, as a general principle, the Commission has acknowledged that significant differences in service characteristics and corresponding cost differences should be reflected in rates. For instance, high-voltage service is usually priced separately from low-voltage service. Interruptible and curtailable service is typically priced lower in recognition of differences in the quality and reliability of the service, as well as cost of service differences.

The recently adopted standby rates provide another example. These rates were carefully

constructed to align with the service characteristics and corresponding differences in the cost of service. The new costing and pricing approach ensures that customers are not paying for a higher quality service than they are receiving. While the calculations are complex, they enable standby customers to pay a fair price for protecting a specific portion of their load against an outage. In turn, these customers are provided with appropriate price signals that ensure their rates are aligned with differences in the cost of providing the actual service they are receiving (not the system average grade of service).

Q. Is it appropriate, from an economic perspective, for prices to reflect differences in service characteristics?

A. Yes. In both regulated and unregulated markets, it is quite common for prices to vary depending on the quality and other characteristics of the item being purchased. A Ferrari is different from a Chevrolet, and it is priced accordingly.

The approach used in setting rates for different voltage levels, peak usage characteristics, interruptible service, and standby service is consistent with a broader philosophical stance taken by the Commission: significant differences in service characteristics should be considered while designing rates. Ideally, the rate design will reasonably reflect cost and quality differences.

This fundamental principle explains many aspects of Con Edison's existing rate design. For instance, high peak demands impose greater costs than lower demands. This difference in service characteristics and the associated cost differences are considered while allocating the revenue requirement and designing rates. Perfect cost alignment is not feasible, and some cost averaging is inevitable. However, the Commission has been steadily moving toward improved alignment of rates and costs.

Q. Should high and low-cost customers continue to be lumped together merely because this is the way rates have always been designed?

A. No. While tariff changes can be controversial, there has been a valuable trend toward the use of increasingly sophisticated analytical methods to understand differences in the service characteristics and the cost of providing service to different types of customers. The Commission has been striving to improve its understanding of the underlying cost differences in order to move rates closer to the actual cost of service, except where that movement would be inconsistent with countervailing policy considerations.

To my knowledge, the Commission has never studied differences in the cost of serving Westchester County relative to New York City. Nor has it studied the cost of serving the Company's network customers relative to its radial customers. Nor has the Commission evaluated options for improving the alignment of

the Company's rates with the cost and service quality being provided to different categories of customers, or between Westchester County and New York City.

Q. Aside from their general desire to be treated fairly, have the Consortium members expressed concerns about the financial strain imposed by Con Edison's rates?

A. Yes. One indication of this concern is that some of the resolutions adopted by Consortium members specifically referenced past increases in Con Edison's rates, in addition to the pending proposed increases. The Commission has also received over 2,000 comments from customers, including a plethora of Westchester residents, concerning the adverse impact of the recent rate hikes on their finances and well-being.

Q. Have you investigated their concern to see if it is valid?

A. Yes. I reviewed the annual rate increases outlined in the Joint Proposals in the Company's

2016, 2019, and 2022 rate proceedings. Schedule 1 of my Exhibit__(WMC-1) summarizes this part of my work. Schedule 1 focuses on electric and Schedule 2 focuses on gas rates.

The Company's residential SC1 electric delivery rates have increased by more than 4% every year since 2016. The increases approached or exceeded 6% in several years. The situation for small business electric customers is also troubling. The small general service SC3 electric delivery rates increased by at least 4% every year during the past decade, and the increases in a few years were even higher.

The year-to-year pattern of gas rate changes was less consistent, but the overall results were at least as disturbing. For instance, the delivery rates now being charged SC1 residential gas customers are roughly 90% higher than the rates they were being charged in 2016. The delivery rates currently being charged to SC3 gas residential (heating) customers are more than

double those in 2016. Small businesses that do not use gas for heating are now paying SC2 gas rates that are 70% higher than the rates they paid in 2016. The companies that use gas heating are paying SC2 rates more than double those in 2016.

Q. There has been a lot of inflation over that period, particularly during 2021 and 2022. Did you check to see if the rates have kept pace with inflation?

A. Yes. I compared the cumulative compound increases from these recent Joint Proposals to the inflation rate over the same periods. Unquestionably, the Company's delivery rates have been increasing faster than the overall inflation rate.

For example, from 2022 to 2025, the SC1 residential electric delivery rates increased approximately 1.8 times as fast as the consumer price index. Over the same period, the residential SC1 gas rates increased 1.9 times

the consumer inflation rate, while residential SC3 gas heating delivery rates increased 3.4 times as fast as the overall increase in consumer prices.

Small business customers have not been spared from the financial impact of rapidly increasing utility rates. Using the same approach to analyze changes in the Company's delivery rates for small general service, I focused on the SC2 electric and gas general service rates. These tariffs apply to most small businesses operating within the Westchester County municipalities. I examined changes in these rates relative to the consumer price index and the GDP deflator, a broader measure of the overall inflation rate.

The SC2 electric delivery rates are more than 20% higher than in 2022. That is more than 2.4 times the projected cumulative change in the GDP deflator through the end of 2025, and more than twice the projected consumer price index through the end of 2025. Increases in the Company's SC3

small general service gas rates also far outstripped the overall inflation rate. SC2 Rate I increased 4.3 times the GDP inflation rate and 4.0 times the CPI inflation rate. SC2 Rate II increased 6.1 times the GDP inflation rate and 5.8 times the CPI inflation rate.

Q. Have you compared the Company's residential delivery rate proposals to the anticipated inflation rate?

A. Yes. The bottom line is clear: if the Company's proposals are accepted, rates will continue to increase faster than the inflation rate. If the proposed SC1 electric rate increases were entirely accepted, rates would increase 5.2 times the change in the Consumer Price Index from 2025 to 2028 (as projected by the Congressional Budget Office). Similarly, the SC1 gas rates would increase 4.1 times the increase in the CPI, and the SC3 (heating) gas rates applicable to single-family homes, duplexes, triplexes, and quadplexes would increase 9.2 times as fast as the CPI.

Q. Have you compared the Company's small general service delivery rate proposals to the overall inflation rate?

A. Yes. As shown in Table 2 of Schedule 1, the SC2 electric rate will increase 5.7 times faster than the increase in the GDP deflator. As shown on Tables 2 and 3 of Schedule 2, the SC2 Rate I gas rate will increase 4.3 times as fast as the GDP inflation rate, while the SC2 Rate II gas heating rate will increase 6.1 times as fast as the GDP inflation rate.

Q. Why have the Company's rates increased faster than the overall inflation rate?

A. Many factors contribute to this pattern, with the impact of each factor varying from case to case. In general, however, it is fair to say that the most significant driving force behind these rate increases has been increases in investments and operating costs that were negotiated with and approved by the DPS Staff. For example, the Company has invested hundreds

of millions of dollars to improve reliability and storm resiliency. Similarly, it has been replacing leak-prone pipes in its gas distribution system. Both of these ambitious efforts offer significant benefits, but are costly endeavors that continue to push up already sky-high rates.

Q. You've occasionally characterized Con Edison's rates as being "high." Is this a fair description, or were you being somewhat hyperbolic?

A. This is an accurate description. As I mentioned, the Company's rates have been consistently increasing faster than the rate of inflation, which is justification enough for describing them as high. But the problem runs deeper than that. The Company charges some of the highest rates I have ever encountered anywhere in the United States. **In fact, the small cities, towns, and villages in Westchester County are paying some of the highest electric rates in the United States.**

Q. Can you provide any evidence to support that statement?

A. Yes. The Energy Information Administration (EIA) publishes a wealth of data on energy usage and costs. One of the statistics they publish monthly is called the Average Price of Electricity to Ultimate Customers by End-Use Sector. This per kWh data is compiled and calculated using information provided by individual utilities to the federal government on Form EIA-826. The EIA analyzes this data by dividing the revenues received from customers by the kWh sales of electrical energy provided to those customers.

Looking at the EIA data for the most recent calendar year (2023), the average price of electricity sold to Residential customers was approximately $.1600 (16 cents), and the average paid by Commercial customers was $.1259. The EIA does not publish price data for individual utilities, but the granular data needed to

estimate prices is available from the EIA. After obtaining this data, I applied a similar approach to the one used by the EIA. The primary conclusion I reached is that the bundled price paid by Con Edison's residential customers is nearly double the average price paid by a typical residential customer in the United States. Similarly, Con Edison's commercial customers pay almost 50% more than the national average.

Q. Is there other evidence that customers in Westchester County are paying higher rates than customers in other, comparable locations?

A. Yes. I compared the Company's rates with those charged by hundreds of individual utilities, using granular data that these utilities have submitted on Form EIA-826. The results of this comparison are summarized on Schedule 4 of Exhibit__(WMC-1). This exhibit is supported by an extensive data set for bundled rates (including the cost of generating electricity)

and a smaller data set for delivery rates (excluding generation costs).

Q. How many entities submit data on Form EIA-826?

A. The 2023 data set has revenue and kWh data from 2,500 entities. This number includes public utilities, Community Choice Aggregators, Retail Power Marketers, and reporting entities that generate electricity at a retail customer's premises Behind the Meter. To improve comparability, I focused on public utilities, including Cooperatives, Municipal utilities, government entities like the Long Island Lighting Company, and investor-owned utilities like Con Edison.

Most of these utilities provided usable data for both residential and commercial customers, but I encountered some redundancies, gaps, and other data complications in a few cases. All told, I was able to estimate bundled residential prices for 1,334 utilities and commercial prices for 1,390 utilities (including Con Edison). I could

estimate residential delivery prices for 25 utilities and commercial delivery prices for 33 utilities. The delivery pricing data is limited because utilities in many states only provide bundled service to residential and commercial accounts.

Q. What did you learn by studying this data?

A. Con Edison's residential rates are higher than those charged by approximately 99.5% of the Cooperatives, 99.1% of the Municipal and other government utilities, and 91.0% of the Investor Owned utilities. All told, approximately 98.1% of all utilities in the United States have lower residential prices than Con Edison, as shown on Schedule 4, Table 1 of Exhibit__(WMC-1).

Similarly, I estimate the prices paid by the Company's commercial customers are higher than those paid by the average commercial customer served by approximately 98.4% of the Cooperatives, 99.2% of the government utilities, and 92.1% of the Investor Owned utilities in the

United States. All told, I estimate that approximately 97.6% of all utilities in the United States have lower commercial rates than Con Edison.

Q. In your opinion, is this evidence that Con Edison has not engineered its system appropriately, or that it is inefficient?

A. No. Throughout my career, I have had the opportunity to study many different aspects of many different public utility systems. This work has included research concerning integrated resource planning, mergers and acquisitions of these systems, changes to their depreciation rates, and public policy efforts to encourage energy efficiency and respond to climate change concerns. This research has provided me with a broad nationwide perspective on the many complex factors explaining why some utilities have high rates while others have lower ones.

I am not asserting that Con Edison is being poorly managed or poorly engineered. That is

not the focus of my testimony. The explanation for Con Edison's high rates lies with many factors, including policy decisions of this Commission and other governmental bodies within its service territory, upward cost pressures, including capital investments negotiated with important constituencies, and (most importantly) the unusual nature of its extremely congested, high density service area in New York City.

It is helpful to realize that every public utility system has a distinct combination of different components, reflecting engineering decisions that are made over a lengthy historical time frame, reflecting the unique characteristics and circumstance of its geographic service area, and the policy guidance and constraints provided by local municipalities and various state agencies.

The cost of building and operating each system varies depending on the underlying characteristics of the service area and the

cost-effectiveness of various decisions made by the utility over many decades. These decisions involve economic tradeoffs that cannot easily be summarized in one or two simple statistics, making it more difficult to pinpoint why the costs incurred in operating one system are higher than those incurred in operating another. In the unique case of New York City, however, it is self-evident that most of the upward cost pressure is closely related to the extraordinary conditions in the largest city in the United States.

Q. Can you provide some insight into why it is hard to pinpoint precisely why Con Edison's rates are higher than those charged by most other utilities?

A. Yes. Consider the cost of generating electricity. This is a significant component of the bundled retail price paid by residential and commercial customers. The structure of energy markets differs across the United States. In New York State, independent power producers are

primarily responsible for electrical generation. In states like Virginia, regulated utilities still operate many generating plants, but a partially deregulated wholesale market exists. In states like Florida, the distribution utilities own and manage nearly all generating plants, and wholesale transactions frequently occur on a peer-to-peer negotiated basis.

In addition to these differences in market structure, the underlying cost conditions can differ significantly from state to state. For instance, low-cost solar energy is not as plentiful in New York as in states like North Carolina and Florida due to differences in average annual solar radiation, available land suitable for utility-scale power production, and other variables.

Similarly, there are differences in energy market conditions that vary state by state. Some of these differences are not stable; they fluctuate month-to-month and year-by-year. Yet,

it is hard to develop precise comparisons because of the effects of long-term contracts, market hedging, and other complications. Still, it is fair to say that utilities in some locations have benefited from opportunities to generate electricity using low-priced natural gas that did not exist in other states during the same period. These opportunities, and the resulting discrepancies in the delivered price of electricity purchased by end-use consumers, cannot easily be summarized. These variations relate to historical circumstances, proximity to interstate pipelines, and other variables that may or may not be within the control of individual utilities operating in each state.

To illustrate this point, data from the EIA suggests generators in New York have recently benefited from lower natural gas prices than their counterparts in Massachusetts. According to the EIA data, the average price of natural gas sold to electric generators in Massachusetts during 2021-2023 was $7.12 per thousand cubic

feet. The analogous average price for gas sold to generators in New York was $4.55, while the price paid by generators in Pennsylvania was $3.81. Meanwhile, generators in Connecticut paid an average of $5.93, and generators in Virginia paid an average of $4.69. The impact of gas price differences on retail electric rates varies, depending on the structure of energy markets in each state, the extent to which generators in that state are relying on natural gas, the extent to which wholesale electric prices equilibrate region-wide, and the extent to which individual generators, utilities and power marketers hedge their exposure to natural gas markets.

Q. Do these factors partially explain why Con Edison's rates are higher than most other utilities?

A. Yes. The average bundled rate per kWh paid by residential customers in the market areas served by the New York Independent System Operator (NYIS) was about $.2203, which was approximately

$.0661 higher than the analogous price of $.1542 paid by residential customers in the United States as a whole, as shown on Schedule 4, Table 1 of Exhibit__(WMC-1). Looking at Table 2 of that same Schedule, you can see a similar situation for the retail bundled price paid by commercial customers of the NYIS utilities. These customers were paying approximately $.0755 more per kWh than the analogous average price paid by bundled commercial customers throughout the United States.

Q. The generation and transmission costs explain about 6 to 8 cents of the difference between Con Edison's rates and the national average. Is that the full explanation?

A. No. The tables I just mentioned confirm that other factors are also involved. Notice that the residential delivery rates of the NYIS utilities were more than two cents higher than the analogous delivery rates, and the commercial delivery rates were approximately one cent higher per kWh than the average in the rest of

the country. This suggests the cost of generation is not the only factor to consider. There are also distribution system-related factors that push Con Edison's prices upward.

Confirmation of this point can be obtained by comparing the rates charged by Con Edison with those charged by other New York Utilities. According to the EIA data, Con Edison's bundled electric rates are about 9 cents higher than those of other state utilities. This discrepancy is attributable primarily to transmission constraints, congestion, and other difficulties that drive up the cost of delivering energy to Con Edison's service area.

There are many unique, costly challenges involved with safely and reliably moving electrical power through an extraordinarily congested metropolitan area. As a result, the Company's delivery rates are significantly higher than those in other parts of New York State. That is why the Company's average

residential delivery rate in 2023 of $.1940 was about five cents higher than the statewide average delivery rate of $.1417. Similarly, the residential delivery rate was about 7 cents more than the US average in 2023.

Q. Do you have any concluding thoughts on this topic?

A. Yes. These different data sets and analyses all point to an important conclusion: customers in Westchester County are paying unusually high rates because Con Edison operates in one of the world's largest, most congested cities. The unique conditions in New York City result in unusually high costs, and these costs are placing a significant financial burden on customers in Westchester County, where the service characteristics and cost of service are more typical of Putnam, Orange, Rockland, and similar counties throughout the United States.

To simplify and drive home this point, it would be fair to say that Westchester County customers

are paying higher rates than their counterparts in Orange and Rockland counties because of their greater exposure to the high costs of providing service to one of the world's largest, most congested cities. Further research would be required to precisely quantify the problem, but customers in the small cities, towns, and villages of Westchester County are paying rates that significantly exceed the actual cost of providing service to these communities. In my professional judgment, based upon the available data, Westchester County rates significantly exceed the actual cost of service. This raises substantial concerns as to the justness and reasonableness of current rates for Westchester residents and businesses.

III. RATE DESIGN AND PUBLIC POLICY

Q. Before delving into other issues, please explain your general approach to rate design.

A. Although rate design is more of an art than a science, it is a crucial element of the regulatory process. The Commission's rate design

decisions impact customers, and they can either advance or undermine the Commission's public policy goals. Accordingly, rate design is not an area where deference should be given to the utility's preferences. Those preferences should be weighed against the concerns raised by other interested parties before arriving at a final decision.

Where changes are warranted (as with distinguishing between the cost of network and radial systems), a “business as usual” approach is inappropriate. If a concern is brought to the Commission’s attention, it should gather evidence and make changes where merited. If those changes have the potential to be disruptive, the appropriate response is to apply gradualism, rather than ignoring the evidence.

Fortunately, this philosophy, including a willingness to follow the evidence and to make even potentially disruptive changes on a gradual, cautious basis, is one that this

Commission and other utility regulatory commissions have widely accepted. Although the pace of change has been inconsistent, throughout my career, I have observed policymakers developing a more accurate understanding of cost differences and a willingness to use price signals to reduce harmful emissions, discourage wasteful or unnecessary energy usage, and promote increased energy efficiency.

To cite one crucial example, there was once a time when most utilities in the United States used declining block rates to encourage the usage of electricity and natural gas. This rate design was popular with utilities because it encouraged profitable growth and enabled utilities to keep adding new capacity at a lower cost per unit than was previously feasible, thanks to technological improvements and economies of scale.

By the 1980s, these advantages had diminished, and many state regulatory commissions realized

that economies of scale were no longer a dominant consideration in designing rates. Even if economies of scale were still present, states began to recognize the importance of energy conservation and other countervailing concerns. This convinced many states to reject a "business as usual" approach. They began to phase out declining block rates, and in some cases, they introduced inclining block rates where they felt that was helpful.

This shift in policy occurred in parallel with an enhanced concern about energy efficiency, which has taken the form of newly adopted legislation and regulations, including the imposition of higher standards on manufacturers, changes to building codes, tax credits, and other measures aimed at achieving these widely accepted policy goals. Other efforts have lagged. In some jurisdictions, like New York State, utilities have resisted changes to their rate designs that would improve the incentives for utility customers to install better

insulation in their homes and businesses, to replace older, less energy-efficient appliances such as water heaters and furnaces, and to invest in higher-efficiency lighting.

While there has recently been some beneficial movement in New York, there continue to be opportunities to improve utility rate designs to encourage customers to conserve energy. Improved price signals can enhance and reinforce other state and federal policies, and these signals can potentially achieve results that cannot easily be accomplished through any other means. For instance, some customers would be deeply offended if a law were adopted that required them to restrict their thermostat to a specific range. Yet, those same customers may be perfectly willing to adjust their thermostat toward a more energy-efficient range as a voluntary response to an appropriate price signal.

Q. You have been talking extensively about the cost of providing utility service. Are you familiar with Con Edison's process for developing its rate design?

A. Yes. The Company emphasizes rate continuity and information extracted from its embedded cost of service (ECOS) studies. These studies were developed using a three-step process.

During the first step, "functionalization," costs are organized around operating functions, such as transmission, distribution, and customer accounting. In the second step, called "classification," costs are grouped into three categories: demand-related, commodity-related, and customer-related. The third step is called "allocation." During this stage, specific data is selected and developed into percentage factors that enable the Company to choose how it wants to spread the overall system costs onto specific rate schedules or groups of customers. Finally, the ECOS results were used to develop

the Company's rate design proposals in conjunction with other information.

Having worked with ECOS studies in numerous other jurisdictions, I can attest that the mechanics of this standard three-step process are well-established and not subject to much controversy. However, the interpretation and application of the cost results are often highly controversial, as are some decisions made before the ECOS study is initiated. For example, the Company has structured its electric ECOS studies to separate customers receiving service on high-voltage lines from those receiving service on low-voltage lines. It has not (but could have) done something similar to separate customers receiving service in low-cost areas from customers receiving service in high-cost areas.

This is just one example of the many judgments that are applied during the preparation and application of ECOS studies. Historically, judgments concerning the most appropriate peak

usage factors have often been highly controversial. However, these controversies have waned in most jurisdictions, as specific approaches have gradually become standardized and accepted by the utilities and most other interested parties. Meanwhile, other judgments continue to be highly controversial. In this proceeding, one of the most contentious aspects of the classification and allocation process is likely to be the treatment of costs that are sometimes identified as "customer-related."

There are several reasons why this contentious issue remains a topic of ongoing debate in New York and some other states, including the disproportionate impact of this treatment on smaller residential and small commercial customers in comparison to larger customers, and conflicts that arise between using a high fixed monthly rate to recover these costs and rate design policies that place a greater emphasis on the volumetric rate elements. Many environmental groups and other intervenors continue to believe

that declining block rates and high fixed rates are inconsistent with efforts to encourage energy efficiency, and they undermine efforts to address climate change concerns.

Q. Can you succinctly elaborate on why customer cost estimates are often disputed?

A. Yes. The primary difficulty lies with costs that do not vary in direct proportion to energy usage, peak usage, or the number of customers. By convention, an ECOS study allocates every cost incurred by the utility into one of these three categories. Hence, controversies arise when costs do not closely conform to these categories.

For instance, the cost of trenching, directional drilling, and some other aspects of installing underground gas distribution mains tends to be heavily influenced by the inherent geographic characteristics of the service territory, including the type of soil and the configuration of the street grid where service will be

provided. Although trenching and drilling costs can vary widely, they are not tightly correlated with, nor attributable to, the volume of energy to be distributed, peak day usage or demand, or the number of customers along the route.

A similar problem arises with the cost of directional drilling and other activities involved in installing underground electric distribution lines and poles used with aerial cables. The time spent on drilling and the number of poles required along a particular route will largely be determined by the configuration of the streets and roads. The cost of drilling and the number of poles are not closely correlated with the number of customers (or even the number of buildings) on those streets.

Dr. James Bonbright aptly described the problem in the classic "Principles of Public Utility Rates," published by Columbia University Press in 1961. He starts his explanation by defining

customer costs as "those operating and capital costs that vary with the number of customers regardless, or almost regardless, of power consumption. Included as a minimum are the costs of metering and billing along with whatever other expenses the Company must incur in taking on another customer."

Dr. Bonbright then explains that many analysts are not satisfied with this narrow construal of customer costs. Instead, they expand the definition well beyond this narrow range: "… the really controversial aspect of customer-cost imputation arises because of the cost analyst's frequent practice of including, not just those costs that can be definitely earmarked as incurred for the benefit of specific customers but also a substantial fraction of the annual maintenance and capital costs of the secondary (low voltage) distribution system—a fraction equal to the estimated annual costs of a hypothetical system of minimum capacity. … Their inclusion among the customer costs is defended

on the ground that, since they vary directly with the area of the distribution system (or else with the lengths of the distribution lines, depending on the type of distribution system), they therefore vary indirectly with the number of customers."

In his text, Dr. Bonbright disagrees with this faulty logic: "What this … cost imputation overlooks, of course, is the very weak correlation between the area (or mileage) of a distribution system and the number of customers served by this system. ... Indeed, if the company's entire service area stays fixed, an increase in number of customers does not necessarily betoken any increase whatever in the costs of a minimum-sized distribution system."

Dr. Bonbright describes the inclusion of these minimum system costs in "customer costs" as "indefensible," while admitting there are problems with placing the minimum system cost into any of the other available categories. He

concludes, "the only defensible answer, in my opinion, is that it belongs to none of them. Instead, [the minimum system] should be recognized as a strictly unallocable portion of total costs. And this is the disposition that it would probably receive in an estimate of long-run marginal costs. But the fully-distributed cost analyst dare not avail himself of this solution, since he is the prisoner of his own assumption that 'the sum of the parts equals the whole'. He is therefore under impelling pressure to 'fudge' his cost apportionments by using the category of customer costs as a dumping ground for costs that he cannot plausibly impute to any of his other cost categories."

Q. Is Dr. Bonbright's assessment relevant to the ECOS estimates developed by the Company in these proceedings?

A. Yes. The methodologies adopted by Con Edison's Gas Rate Panel and its Electric Demand Analysis and COS Panel suffer from the flaw described by Dr. Bonbright. This same flaw has been noticed

and emphasized by many experts during the past 60 years. The customer costs calculated by the Company include a comprehensive list of investments and operating expenses that remain unchanged when an additional customer joins the system and when a customer leaves the system. Many of these so-called customer costs do not vary significantly with the number of customers in any given section of the system. Other costs, such as the investment in service lines, may be loosely correlated with the number of customers, but this does not mean that an additional service line must be installed whenever a new customer joins the system, nor is the number of service lines equal to the number of customers.

As a practical matter, allocating minimum system costs and other loosely correlated costs on a weighted per-customer basis often puts upward pressure on the rates paid by smaller customers compared to larger ones. Even if the expenses are "fixed" with respect to peak demand or energy usage, that does not mean they should be

allocated or recovered on a weighted per-customer basis.

Most of these so-called customer costs would not be avoided if the number of customers were smaller, nor would they increase if the number of customers were to rise. For instance, consider the example of a large tenant moving out, and the landlord subsequently renting the space to multiple smaller tenants. The number of customers sharply increases, yet the minimum system costs remain unchanged. Over the long run, even the costs of some items that are physically located at or adjacent to the customer premises (which are typically classified as "customer-related") may vary with expected demand, peak usage, and energy consumption as much as, or more than, the expected number of customers at that location.

There is no sound basis for insisting that these costs must be allocated or recovered on a

weighted-per-customer basis, whether within classes or between classes.

Suppose gas is going to be made available to a particular city block. In that case, the minimum cost of a distribution main will be about the same whether a few large customers connect to the system or dozens of smaller ones. The cost is primarily a function of the length of the street. Similarly, suppose gas will be made available to a newly constructed building. The service line connecting the building to the main will not be proportional to the number of tenants. The primary consideration in sizing the service line will be whether the building is heated with gas heat and how well it is insulated.

The absence of a strong correlation between so-called customer costs and the number of customers (alluded to by Dr. Bonbright) can be most easily visualized in a context where plans are developed to extend a gas distribution

system into a newly developed neighborhood. The minimum cost of the planned distribution system will depend on the geographic characteristics of the area where gas service is going to be made available, rather than the number of customers that will connect to that system.

A similar disconnect between minimum system costs and the number of customers can also be observed with reductions in the number of gas customers. If concerns about methane leakage and climate change were to persuade half of Con Edison's gas customers in a residential neighborhood to install heat pumps and disconnect from the gas system, the total minimum system costs in that area would not decline proportionally with the weighted number of customers. In turn, the calculated minimum system cost per gas customer could potentially double for the remaining customers.

Additional confirmation of this point can be found in the practical difficulties when

customers are encouraged to transition away from natural gas. Suppose some buildings cannot be fully electrified due to non-combustion industrial processes or the impracticality of retrofitting certain older buildings. In that case, the minimum system costs in that area will not disappear or decline proportionally to the reduction in the number of customers. The result will be a substantial increase in the calculated monthly cost per customer as customers leave the system. This thought experiment confirms a key point: the total minimum system costs do not vary in proportion to the number of customers.

IV. CON EDISON RATE DESIGN

Q. Have you reviewed Con Edison's ECOS studies and rate design proposals?

A. Yes. I reviewed the testimony submitted by Con Edison's Gas Rate Panel, its Electric Demand Analysis and COS Panel, and its Electric Rate Panel, along with the workpapers developed by these panels.

Q. Have you prepared some alternatives for the Commission's consideration?

A. Yes. I developed some ECOS calculations and some rate design calculations. A good starting point is to explain some "customer cost" calculations I developed for illustrative purposes.

The first column of Schedule 1 of Exhibit___WMC-2 (electric) and Exhibit___WMC-3 (gas) shows the narrowest interpretation of what has traditionally been called the "basic customer" method. This column includes the costs of billing and receiving payments and metering costs, consistent with Dr. Bonbright's comments. The gas example also includes costs associated with house regulators. On average, this approach yields a monthly cost estimate of less than $5 for residential and small commercial electric customers and less than $4 for the smallest gas customers.

Despite the narrow definition, the cost estimates are not perfectly uniform. They vary with the customer's size. For instance, metering costs can be higher for larger customers, as suggested by comparing the figures listed in the first column of Exhibit WMC-2 (electric). A similar phenomenon is observed in the first column of Exhibit___WMC-3 (gas). I would note that the latter differences also reflect variations in the size and cost of house regulators, which are grouped with meter costs in the Company's ECOS study. Street lighting and signaling customers have the lowest estimated costs because they do not require meters.

The second column adds an estimate of customer accounting and collection costs. Some analysts prefer to include these costs in the analysis, since customer accounting and collection efforts vary with the number of customers. Other analysts exclude some or all of these expenses because those efforts can also vary in magnitude depending on the size of the bill. For example,

late payments, arrearages, and associated collection efforts frequently arise when customers experience difficulty paying their utility bills by the due date. In some cases, the collection problem can be attributed to unusually hot or cold weather, which leads to an unexpected surge in energy consumption or peak usage that was not anticipated in the customer's monthly budget. In other cases, the initial cause may be the loss of employment, a sickness, or another source of financial difficulties. Still, the larger the utility bill, the more likely the financially stressed customer will have trouble paying their utility bill on time. Due to these complicating factors, accounting and collection costs are not homogeneous. Since these cost variations are positively correlated with energy consumption, some cost analysts develop an estimate of customer costs that excludes the portion of accounting and collection costs that varies with bill size.

The third column is much more expansive, including all of the so-called customer costs identified by the Company, except the minimum system costs. These costs include some costs related to electric distribution lines, transformers, and gas distribution mains. Finally, the fourth column in both exhibits shows the approach adopted by the Company, which is one of the broadest definitions of customer costs I have encountered.

Q. Can you briefly comment on the significance of these cost estimates?

A. Yes. To clarify, I am not suggesting that there is only one "right" answer. I believe the Commission can glean valuable insights from each of these cost estimates. There is no need to settle upon a single "best" cost estimate.

The "basic customer approach" yields cost estimates ranging from approximately $5 to $10 per month for residential and small commercial electric customers. The "full cost" approach

ranges from $23 to $36 monthly for residential customers, and approximately $38 to $66 monthly for small commercial customers.

Moreover, considerable cost-averaging occurs within these estimates. If one were to drill down and analyze customer costs in more granular detail, one would undoubtedly find some smaller residential heating customers with costs well below the class average developed by the Company (closer to the estimated $34 monthly cost of the average SC1 gas customer). Conversely, there are undoubtedly some large customers in the newly defined SC3 Rate II category with full costs above the calculated class average.

A 400 square foot apartment near the center of a town in Westchester is likely to have lower calculated costs under the Company's methodology than a 4,000 square foot single-family house on a large lot at the edge of town, because of the need for a longer, more costly service line. Similarly, average costs will vary between

different boroughs and between these boroughs and the communities in Westchester County.

The same principles apply to both electric and gas customers. A tiny bodega receiving electric service under the SC2 tariff will impose full costs below the calculated class average of $66 per month. While it is hard to predict the extent of the discrepancy, an accurately developed cost estimate for a very small commercial customer could be less than the $36 amount the Company calculated for its SC1 electric customers, as shown at the top of the fourth column on Exhibit__(WMC-2). The key point is that the ECOS results should not be interpreted as precise estimates of the actual cost to serve very small or large customers, since they are based upon the average characteristics of a customer class that is not homogeneous.

Q. Is it practical to avoid this problem?

A. Yes. Customer classes can have varying levels of

homogeneity. To cite a simple example, Orange and Rockland has a few, broadly defined customer classes. This results in greater class heterogeneity than the approach taken by Con Edison. However, even Con Edison's customer classes can be further disaggregated to improve homogeneity. For instance, smaller customers can be separated from larger ones. Similarly, customers in small towns and cities can be separated from customers in larger cities, or network customers can be separated from radial customers to improve the homogeneity of each class. This would make developing more granular and precise embedded cost results feasible.

If customers on the predominantly radial systems in uncongested small towns and cities were separated from the customers in congested cities, the cost information used in developing rates would more accurately reflect the actual cost of serving the characteristics of each service area or homogeneous group of customers. By separating low-cost customers from high-cost

customers, rates can be more closely aligned with the relevant cost characteristics, and better, more precise price signals can be sent to all customers.

Q. Can you point to evidence demonstrating the effect of grouping and averaging customers with heterogeneous cost characteristics?

A. Yes. In this proceeding, for the first time, the Company studied the cost of serving residential gas customers based upon the number of dwelling units behind the meter. A somewhat analogous disaggregation has long been a feature of the Company's electric ECOS studies, but not its gas studies. The last three rows of Schedule 1 of Exhibit___WMC-3 confirm the impact of averaging costs across disparate types of customers.

Customers with larger buildings use more gas, impose higher peak demands, and require more costly facilities to connect the distribution system to the customer's premises. Because the SC3 class averages costs for various sizes and

types of customers in widely different geographic areas, these differences are not ordinarily revealed in the ECOS study results. Consider, for example, the minimum system required to provide gas service to an apartment building in a municipality in Westchester County compared to the cost of providing service to a comparable-sized building in New York City.

As discussed earlier in my testimony, there are significant cost differences depending on the geographic location of customers; yet, these differences are ignored in the Company's ECOS studies. Similarly, there are differences between the cost of serving 10 individually metered apartments and a group of 10 identical apartments located behind a single meter. However, the Company's ECOS studies do not analyze these differences in detail. As a result, their cost estimates tend to overstate the minimum system costs required to serve the individually metered residences relative to the master-metered ones.

With class-wide averaging, the cost of billing residential heating customers is approximately $11 per month, including the cost of meters and house regulators. In this proceeding, for the first time, the Company analyzed SC3 gas customers with 1 to 4 dwelling units separately from those with five or more dwelling units behind the meter. After disaggregation, monthly billing and metering costs dropped to less than $7 for smaller customers. With deaveraging, the calculated cost for the larger customers increased to approximately $67 per month.

This disaggregation exercise quantified for the first time some of the long-standing disparities in the billing and metering costs that exist within this class. The average billing and metering costs for customers with five or more units behind the meter are nearly 10 times higher than the average for the remaining, smaller customers. Splitting the SC3 class reduces the extent of cost averaging and

achieves a higher degree of class-specific homogeneity.

Moreover, cost disparities will continue within the SC3 class, even if a distinction is drawn based on the number of units behind the meter. To illustrate this point, consider a simple example: some 10-unit, 6,000 square-foot apartment buildings are connected to a particular gas distribution line. A nearby 6,000-square-foot single-family home is connected to the same distribution main. All the buildings are heated with natural gas. All of them have identical service lines connecting them to the distribution system. The calculated minimum system cost, developed in the Company's ECOS study, is assumed to be approximately $34 per month for each apartment if they are individually metered, and $348 per month for the entire apartment building if all 10 units are behind the meter (heating is included in the rent). So far, so good. However, this example's large 6,000 square-foot single-family house

remains an anomaly. The SC3 Rate I average cost of $34 per month is applied to this customer; however, the actual cost is approximately $348 (the same as any other building with the same characteristics).

Cost disparities due to differences in the service line, house regulator, and variations in the number of customers and dwelling units are not precisely analyzed in the Company's ECOS studies. While the new study is an improvement, some disparities continue to be ignored, and some customers are assigned an average share of the class-wide calculated costs, despite having different cost characteristics than the remainder of the customer class.

Q. What conclusions do you draw from this analysis?

A. First, ECOS studies are a helpful information source when considering the appropriate rate design. However, they should not be treated as perfect reflections of the underlying cost conditions. This is particularly clear when

significant heterogeneity exists within a service class. A good example of this problem is the failure to study cost differences between network and radial systems.

Second, **significant cost differences related to the customer's location have been largely overlooked in the Company's ECOS studies.** Locations like midtown Manhattan are more costly to serve than others; these geographic differences have not been analyzed in depth.

Third, differences in the cost of facilities at or adjacent to the customer's premises have not been thoroughly analyzed. Due to intraclass averaging, small customers are assumed to incur the same service line and house regulator costs as larger customers. However, this is not a valid assumption. With further disaggregation and analysis, the estimated cost of serving the smallest customers would be lower, and the cost of serving the largest customers would be higher than the class-wide averages developed in the

Company's ECOS studies. While this problem cannot be eliminated, it can be reduced by increasing class homogeneity.

Fourth, the calculated level of customer costs developed by the Company falls toward one end of the range of alternatives. Other approaches yield significantly lower monthly customer costs.

Fifth, the Commission should balance its interest in having the rate design reflect underlying cost patterns and its interest in developing price signals that support its other policy objectives.

V. RECOMMENDATIONS

Q. To start our discussion of your recommendations, can you briefly explain how you think the Commission should respond to the Company's rate design proposals?

A. Yes. My primary suggestion is for the Commission to avoid high customer charges (or minimum

bills) and low tail block rates. This will **improve the price signals** provided in the Company's rate design. Together with customer engagement efforts, gradually increasing tail block rates will empower and encourage customers to take greater control over their utility bills and more clearly and effectively reward them for investing in more insulation and energy-efficient appliances, heat pumps, and heating systems. Improved price signals will encourage customers to make lifestyle adjustments that help them use energy more efficiently (e.g., using automated thermostats to adjust temperatures for maximum efficiency while maintaining comfort).

Q. Can you briefly describe Con Edison's proposals for its electric customer charges and volumetric rates?

A. Yes. The Company proposes increasing the SC1 electric residential and religious customer charge by 15%, from $20.00 to $23.00 per month. It proposes to increase the block rates by

approximately 20%. The proposed changes are similar for the SC2 rate, which applies to small general service customers. A similar approach was used with most of the electric tariffs.

Q. Can you briefly describe Con Edison's current and proposed minimum charges and volumetric rates for gas customers?

A. Yes. The company proposes increasing the minimum bill and the energy rate in the SC1 residential non-heating class by approximately 16.5%. For SC2 Rate 1 (small general service non-heating), it proposes increasing the minimum bill by 21.3%, which exceeds the proposed increases to the energy rates of 9.1% in the initial block, 15.3% in the middle block, and 19.7% in the tail block. The Company proposes increasing the minimum bill for SC2 Rate 2 (small general service heating) customers by 21.3%. This exceeds the proposed increase to the energy rates of 14.8% in the initial block and 18.5% in the middle block. It proposes to increase the tail block by 24.3%.

More dramatic changes are proposed for customers in the SC3 residential heating class. The Company is proposing to split this class into two. Customers with one to four dwelling units behind the meter would be placed in Rate I, while customers with five or more dwelling units behind the meter would be placed in Rate II. The minimum bill paid by Rate I customers would increase by 25.0% while the minimum bill paid by Rate II customers would increase by approximately 87.5%. Both groups of customers would also face substantial increases in the energy rate. For Rate I, the initial block would increase by 26.1%, the middle block would increase by 32.1%, and the tail block would increase by 38.6% if the Company's proposals are accepted. Rate II customers would experience an increase in the initial block of 11.0%, while the middle block would increase by 13.6%, and the tail block would increase by 16.3%.

Q. It is sometimes argued that rates should be closely aligned with "customer costs," including an allocated share of minimum system costs. Do you agree?

A. No. For one thing, other legitimate ways to estimate customer costs yield significantly lower estimates. However, aside from that controversy, it is essential to realize that tradeoffs are involved in developing an optimal rate design. There is no compelling reason to recover the cost of a minimum-size distribution system, or the fixed costs of facilities located at or near the customer's premises, through the electric customer charge or gas minimum bill. Instead, these costs can be recovered through the demand and volumetric charges. That pricing strategy has several advantages, the most prominent one being that it provides stronger price signals to encourage customers to conserve energy and limit their peak usage. To help promote the Commission's policy goals, it is preferable to recover costs on a volumetric

basis, rather than through a fixed monthly charge.

Q. Have you developed some exhibits that help clarify your recommendations?

A. Yes. I developed four sets of illustrative rates. Schedule 2 of EXHIBIT___(WMC-2) is based upon the Company's proposed electric revenue requirement, while EXHIBIT___(WMC-2), Schedule 4 assumes a lower revenue requirement. EXHIBIT___(WMC-3), Schedule 2 uses the Company's proposed gas revenue requirement, while EXHIBIT___(WMC-4), Schedule 4 assumes a lower revenue requirement.

Q. Can you briefly explain the differences between your illustrative rates and those proposed by the Company?

A. Yes. **I recommend constraining increases to the fixed monthly rate elements to improve price signals**, encouraging customers to use energy efficiently, and helping advance the state's policy goals related to climate change. More

specifically, unlike the Company, I've consistently increased the customer charge and minimum bill rate elements by less than the volumetric rates. I agree with the Company's proposal to phase out the declining block structure in its gas tariffs. Still, I recommend moving moderately faster than the Company has proposed. Similarly, I recommend **slightly steepening the inclining block rate element** in the Company's summer residential electric rates, **and** slightly increasing the tail block winter rate element relative to the initial block, thereby **introducing an inclining block structure into the winter residential electric rate design.**

Gradually transitioning rates in this direction will provide stronger price signals to encourage energy efficiency and reduce peak demand. This pricing approach is more consistent with the state's policy goals because it reinforces and strengthens the impact of outreach and customer education efforts and financial incentives that

encourage customers to improve their weatherization and install high-efficiency heating and cooling systems.

Q. What is your recommendation concerning the Company's proposal to disaggregate the SC3 gas heating class based upon the number of dwelling units behind the meter?

A. I see merit in the Company's proposal, but I have two concerns that prevent me from firmly endorsing it. First, the Company's proposal will significantly increase the rates paid by some of these customers. If this class is restructured, the resulting rate changes should be introduced gradually to avoid excessive rate impacts. Second, the proposed restructuring may not be optimal. Further research and analysis should be conducted to determine whether this is the best way to organize SC3 customers into subgroups.

From a theoretical perspective, a restructuring of this type is desirable if it moves toward a rate design in which all customers in each class

have relatively uniform cost characteristics. This ensures that the extent of arbitrary cost averaging within each group is minimized, making improved price signals more feasible.

To achieve an optimal, yet practical, degree of intraclass homogeneity, it might be preferable to place customers with normal-sized single dwelling units on a separate rate, rather than grouping them with larger multi-dwelling customers like triplexes and quadplexes. Further analysis is needed to determine whether it would be preferable to group duplexes, triplexes, and quadplexes with larger multi-dwelling customers or whether a third group should be established between ordinary single-family customers and the larger multifamily customers. If three rate groups are established, additional research will be needed to determine the most appropriate breaking point. The Company has drawn a line between four and five units, which might be optimal. More thought needs to be given to the tradeoffs involved with this change in the

Company's rate design, to determine whether a different dividing line would be preferable.

I have one more concern, which causes me to hold back from firmly endorsing the proposed change. Additional research should be conducted to determine whether greater class homogeneity is achievable. Single-family residences can range from less than 500 square feet to more than 5,000 square feet. Given this heterogeneity, it might be appropriate to consider other factors besides the number of dwelling units behind the meter.

Further analysis would be beneficial to determine the best way to restructure the SC3 class before adopting permanent changes to this part of the Company's tariff structure. The Commission should strive to ensure that the revised rate structure is a clear improvement over the existing one, rather than an arbitrary one.

One straightforward method for accomplishing greater homogeneity would be to group the largest single-family residences with multi-family residential customers. A simple way to implement this would be to rely on a three-year moving average of gas usage. A computerized analysis of historical billing data would identify outliers (residential customers using inordinately large volumes of energy). They could be automatically moved to the multi-family rate, or contacted and offered this option. Either approach would be viable. The goal would be to improve class homogeneity by grouping unusual single-family residences with small- to mid-sized multi-dwelling customers having similar cost characteristics.

One final point: If the Commission is convinced that the proposed disaggregation of the SC3 residential gas heating rate is moving in the right direction, the **rate modifications should be introduced gradually** to avoid unreasonably large rate impacts. Gradualism will also

simplify the rebalancing process by leaving flexibility for future improvements without imposing excessive fluctuations in customer bills.

Q. Do you have any other recommendations concerning the Company's rate design?

A. Yes. While further research is needed, the evidence I have put forward in my testimony is sufficient to justify a first step: refining the Company's rate design to distinguish between network and radial customers and acknowledging the inherent unfairness of Westchester customers paying rates well above their cost of service.

Radial customers in **Westchester County are paying extraordinarily high rates** for electric delivery service. These high rates result from high costs incurred in serving network customers, not the cost of serving radial customers. Excessive cost averaging is causing these customers to pay substantially more than similarly situated customers in the adjacent

service areas of Orange & Rockland, Central Hudson, and Long Island Lighting Company. The Commission can start rectifying this longstanding inequity by applying a slight discount to the rates paid by radial customers in this case. Further, **the Commission should require the Company to study the problem of excessive rate averaging between Westchester County and New York City**, and to provide the Commission with options for rectifying the existing inequities, such as creation of a separate rate district, revisions to revenue allocations or the introduction of innovative rate designs that better account for differences in service quality.

Applying **a small discount on the rates paid by radial customers would be a reasonable first step** toward alleviating the burden on Westchester County customers who are burdened with an unreasonable share of the high costs incurred in the highest cost, most congested portions of the Company's service areas. After

additional research has been conducted, the Commission can determine whether, and to what extent, this discount should be expanded and what other steps should be taken to more fully reflect differences in the cost and quality of service provided to different types of customers.

Q. What is your recommendation concerning the size of the discount to be adopted in this proceeding?

A. The magnitude of the discount could vary, depending on the final revenue requirement adopted by the Commission. However, a discount of about 5% (applied to the rates paid by radial customers) would be reasonable. A discount in this range would achieve a modest rebalancing of rates, thereby ameliorating some of the existing inequities, without imposing unreasonably large bill impacts on network customers. Since there are substantially more network customers than radial customers, the corresponding increase in

the rates paid by network customers should be less than 1%.

Q. Should the fact that recent rate increases exceed the inflation rate be a factor in establishing rates?

A. Yes. The ability to pay and economic conditions have traditionally played an important role in the review of rate increase requests by utility regulators. Given the increasing number of customers receiving targeted assistance through special programs, the increasing amount of uncollectibles and the adverse impact of sky-high utility rates on the lower middle class and those with fixed incomes, targeted low income rates are not a sufficient solution to high utility rates. Thus, **the Municipal Consortium strongly urges the Commission to** reject low priority, discretionary projects in order to **severely limit any increase** while maintaining safe and reliable service.

Q. Does this conclude your direct testimony filed with the Commission on May 30, 2025?

A. Yes.